Yoonhyuk Choi (Last updated: Nov. 2026)

Assistant Professor
Sookmyung Women's University
Department of Artificial Intelligence

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Research Interest

My research interests include **Machine Learning** and **Data Mining**, particularly *Large Language Models*, *Graph Neural Networks*, and *Recommender Systems*.

Education

•Seoul National University Mar. 2019 - Aug. 2023
Ph.D. & M.S., Computer Engineering Advisor: Chong-Kwon Kim

•University of Seoul Mar. 2013 - Feb. 2019
B.S., Computer Science Advisor: Eui-Kyeong Hong

Work Experience

•Assistant Professor Sep. 2025 -
Sookmyung Women's University, Seoul, South Korea
Department of AI

•Research scientist Oct. 2024 - Aug. 2025
Samsung SDS, Seoul, South Korea
Preceding AI Lab (LLM, RAG)

•Postdoc. associate Nov. 2023 - Sep. 2024
Arizona State University, Tempe, United States (PI: Selcuk Candan and Huan Liu)
School of Computing and Augmented Intelligence (SCAI)

•Postdoc. associate Sep. 2023 - Nov. 2023
Korea Institute of Energy Technology, Naju, South Korea (PI: Chong-Kwon Kim)
Energy AI

•Backend engineer Jun. 2018 - Sep. 2018
nTOPAZ, Seoul, South Korea
Back-end Engineer

Publications (Google Scholar) C: conference, J: journal

- **(C29)** Syndrome-Guided Edge-Relation Decoding for Heterophilic Graph Mining (link)
  Yoonhyuk Choi, Jiho Choi, Chaewon Sung, Yumin Lee, Hyunseo Oh, Eunseo Back, Chanran Kim, Yeowon Jeon, Chaeyeon Lee, JongW
  WSDM '27 (Main)

- **(C28)** A Differentiable 3D Scene Graph Metric via Contextual Hellinger Triplet Geometry (link)
  U-Chae Jun, **Yoonhyuk Choi***, and Jiwoo Kang*
  NeurIPS '26

- **(J27)** Region-Transferable Adaptive Graph Learning for Potential Wetland Prioritization (link)
  Yoonhyuk Choi, and Chong-Kwon Kim*
  Scientific Reports '26 (IF: 4.9)

- **(J26)** Dynamic Sequential Recommendation via Lag-Aware Graph Learning (link)
  Yoonhyuk Choi, Jiho Choi, Hawon Shin, Yumin Lee, Chanran Kim, and Chong-Kwon Kim*
  Information Sciences '26

- **(C25)** SWORD: Scene-Grounded Wasserstein Object-Relation Distance for 3D Scene Graph Prediction (link)
  U-Chae Jun, Jaeeun Ko, **Yoonhyuk Choi***, Jiwoo Kang*
  CIKM '26

- **(C24)** From Oracle Gains to Repair Priorities: Conservative Split-Conditioned Diagnosis for Modular RAG (link)
  Chaewon Sung, Chong-Kwon Kim*, **Yoonhyuk Choi***
  CIKM '26 (short)

- **(C23)** Gauge-Equivariant Graph Networks via Self-Interference Cancellation (link)
  Yoonhyuk Choi, Jiho Choi, Jiwoo Kang*
  ICML '26

- **(J22)** Uncertainty-Aware Adaptive Edge Blocking in Heterophilic Graph Neural Networks (link)
  Jiho Choi, Yumin Lee, Chanran Kim, Hawon Shin, Chong-Kwon Kim, **Yoonhyuk Choi***
  ACM TIST '26 (IF: 10.7)

- **(J21)** Edge-conditioned Markov Kernels for Stable Label Refinement on Heterophilous Graphs (link)
  Taewook Ko, **Yoonhyuk Choi***
  Information Sciences '26 (IF: 8.2)

- **(C20)** Delay-Aware Sequential Recommendation with Dynamic Graphs and Variate-Temporal Decomposition (link)
  Yoonhyuk Choi
  SIGIR '26 (short)

- **(C19)** Sheaf Graph Neural Networks via PAC-Bayes Spectral Optimization (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Jongwook Kim, Chong-Kwon Kim*
  AAAI '26

- **(J18)** Identifying Heterophilic Neighbors via Confidence-based Subgraph Matching for Graph Neural Networks (link)
  Yoonhyuk Choi, Chong-Kwon Kim*
  Artificial Intelligence '26 (IF: 4.6)

- **(J17)** LLM-Guided Review Condensation for Text-Convolutional Recommendation (link)
  Yoonhyuk Choi, Chong-Kwon Kim*
  IEEE Access '26 (IF: 3.6)

- **(J16)** Beyond Binary: Improving Signed Message Passing in Graph Neural Networks for Multi-Class Graphs (link)
  Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim*
  IEEE TPAMI '25 (IF: 20.8)

- **(J15)** Hierarchical Hyperbolic Embeddings for Review-Driven Cross-Domain Recommendation (link)
  Yoonhyuk Choi, Chong-Kwon Kim*
  IEEE Access '25 (IF: 3.6)

- **(J14)** Generalization of Knowledge Transfer with User Reviews for Cross-Domain Recommendation (link)
  Yoonhyuk Choi, Chong-Kwon Kim*
  IEEE Access '25 (IF: 3.6)

- **(C13)** Selective Blocking for Message-Passing Neural Networks on Heterophilic Graphs (link)
  Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim*
  UAI '25

- **(C12)** Review-Based Hyperbolic Cross-Domain Recommendation (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim*
  WSDM '25

- **(C11)** Mitigating Overfitting in Graph Neural Networks via Feature and Hyperplane Perturbation (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim*
  WSDM '25

- **(J10)** Beyond Message-Passing: Generalization of Graph Neural Networks via Feature Perturbation for ... (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim*
  IEEE TNNLS '24 (IF: 14.25)

- **(C9)** Improving the Text Convolution Mechanism with Large Language Model for ... (link)
  Yoonhyuk Choi, Fahim Tasneema Azad
  IEEE Big Data '24 (short)

- **(C8)** Prioritizing Potential Wetland Areas via Region-to-Region Knowledge Transfer and Adaptive Propagation (link)
  Yoonhyuk Choi, Reepal Shah, John Sabo, Selcuk Candan*, Huan Liu*
  IEEE Big Data '24

- **(C7)** Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning (link)
  Ahmet Kapkiç, Pratanu Mandal, Shu Wan, Paras Sheth, Abhinav Gorantla, **Yoonhyuk Choi**, Huan Liu*, K Selçuk Candan*
  CIKM '24 (benchmark)

- **(C6)** Universal Graph Contrastive Learning with a Novel Laplacian Perturbation (link)
  Taewook Ko, **Yoonhyuk Choi**, Chong-Kwon Kim*
  UAI '23

- **(J5)** A spectral graph convolution for signed directed graphs via magnetic laplacian (link)
  Taewook Ko, **Yoonhyuk Choi**, Chong-Kwon Kim*
  Neural Networks '23 (IF: 7.8)

- **(J4)** Aspect-oriented unsupervised social link inference on user trajectory data (link)
  Hyungho Byun, **Yoonhyuk Choi**, Chong-Kwon Kim*
  Information Sciences '23 (IF: 8.2)

- **(C3)** Review-Based Domain Disentanglement without Duplicate Users or Contexts for ... (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim*
  CIKM '22

- **(C2)** Finding Heterophilic Neighbors via Confidence-based Subgraph Matching for ... (link)
  Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim*
  CIKM '22

- **(J1)** Dynamic graph convolutional networks with attention mechanism for rumor detection on social media (link)
  Jiho Choi, Taewook Ko, **Yoonhyuk Choi**, Hyungho Byun, Chong-Kwon Kim*
  PLOS ONE '21 (IF: 2.9)

Invited Talks

•Co-Week Academy Jun. 2026
From conventional AI to LLM (PyeongChang, South Korea)

•DGIST Startup Camp Nov. 2025
Connecting the dots (Seoul, South Korea)

•N-EWN Partner Symposium May. 2024
Identifying Potential Sites for Wetlands (St. Augustine, Florida)

Research Projects

•Training Co-Evolving Human–AI to Lead On-Premise Generative Model in Financial Domain 2026-2029
IITP Korea

– Orchestration, Distillation, and Graph RAG

•Development of Reliable, Personalized Clinical Decision-Support Technology 2026-2029
NRF Korea

– LLM and Graph RAG

•Enhancing Large Language Model with RAG 2025
Research project, Samsung SDS

– Personalized RAG and knowledge graph construction on Knox Systems
– Knowledge graph construction and RAG for chunk retrieval
– Tiny-LLM (e.g., Llama 3B) distillation with huge-LLM (e.g., Llama 70B)

•Can Large Language Model Improve the Text Convolution for Review-Based Recommendation? 2024
Research project, Emit Lab

– Integrated the large language model (e.g., Llama 2, GPT-4) with the text convolution algorithm
– Investigated whether applying summarization based on large language models first, instead of performing 2D convolution on the entire text, results in performance improvement
– Used online shopping mall datasets like Amazon and Walmart

•Selection Criteria and Assigned Weightage for Identifying Potential Locations Wetland 2024
Research project funded by NSF (in collaboration with Tulane University)

– Suggested knowledge transfer between different regions and adaptive propagation between grids
– Demonstrated the effectiveness of the framework through real-world scenario
– Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets

•Causal Discovery of Agricultural Mgmt and Reservoir Op. Induced Water Quality Change 2023
Research project funded by NSF (in collaboration with University of Arkansas)

– Developed causal discovery algorithm for water quality improvement and reservoir management
– Considered spatial and temporal variations and validated the causal learning ability
– Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets

•Tracking footprints with graph neural networks for the reduction of virus spread 2021 - 2022
Coursework project, R&D in AI industry

– Suggested spatial-temporal analysis for the next POI prediction to reduce virus spread
– Selected as social contributing project
– Used datasets are Coronamap of South Korea, Gowalla for POI prediction

•Personalized recommendation based on the user's purchasing histories and social network 2020
Industry project funded by Samsung Research

– Introduced time series analysis of users' purchasing history for personalized advertising
– Applied graph neural networks with binary recommendation techniques
– Used customer datasets provided by Samsung Research

•Next POI prediction based on user movements collected through large-scale sensors 2019
Research project funded by Samsung Electronics

– Recommending the next place based on where students visited within Seoul National University
– Developed energy-saving and effective multi-hop transmission technologies for sensor
– Collected datasets by attaching special stickers to participants

Extracurricular Activities

•PC Member 2026
NeurIPS / ICLR / IJCAI / ICML / AAAI / KDD (Aug. track) / WSDM / TheWebConf / TPAMI

•PC Member 2025
ICLR / IJCAI / ICML / KDD (Feb. track) / CIKM / WSDM / TheWebConf

•PC Member 2024
LoG / CIKM / MM

•Reviewer 2023
Journal of IEEE Multimedia / Plos one

•Research Assistant (RA), Graduate Mar. 2019 - Jun. 2021
Funded by Samsung Research

•Teaching Assistant (TA) Mar. 2020 - Jun. 2020
Topic: Social Network Analysis and Anomaly Detection (Advisor: Chong-Kwon Kim)

•Research Assistant (RA), Undergrad Jun. 2017 - Sep. 2017
Distributed Computing Lab (Supervisor: Jin-Suk Kim)

Awards & Grants

•Best Ph.D. Dissertation Award 2023
Seoul National University

•Overseas Short-term Training Scholarship 2023
Chonnam National University

•BK21 Colloquium Graduate Student Fellowship 2023
Seoul National University

•BK21 Star Student Researcher Fellowship 2023
Seoul National University

•SIGIR Travel Awards 2022
For ACM Student Authors with Accepted Long Paper

•BK21 Scholarship 2020
(Graduate) Seoul National University

•Merit-based Scholarship 2018 - 2019
(Undergrad) University of Seoul

Skills

•Languages: Python, C, HTML/CSS

•Tools / Frameworks: PyTorch, torch-geometric, Scikit-learn, Git, Django, AWS, LaTex

References

•ChongKwon Kim: ckim@kentech.ac.kr

•TaeKyung Kwon: tkkwon98@gmail.com

•U Kang: ukang@snu.ac.kr